

Learning to Reset: Enhancing Mathematical Reasoning in LLMs via Autonomous Context Management

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1 Introduction and Related Work

1.1 Introduction

Recent advancements in Language Models (LMs) have demonstrated significant capabilities in mathematical reasoning tasks. However, these models face inherent limitations when their context becomes cluttered with irrelevant information, leading to chaotic reasoning traces and degraded performance. Existing approaches often rely on external memory systems or prompting strategies, but these do not empower the model with the intrinsic ability to manage its own reasoning flow.

This project proposes a framework to address this gap by teaching models to explicitly clean their context through learned behavior. We introduce a special `<clean>` token that signals a context reset, acting as a lightweight mechanism for explicit backtracking. By combining Supervised Fine-Tuning (SFT) on curated expert traces with a modified REINFORCE Leave-One-Out (RLOO) objective, we aim to train models to recognize unproductive reasoning paths and autonomously reset their context to improve accuracy on complex problems.

1.2 Related Work

LLM and Memory Systems: Prior work has explored dynamic context management via external memory systems, such as graph-based memories or hierarchical structures like MemGPT. However, these systems often rely on complex retrieval mechanisms rather than training the model to manage its own immediate context.

Context Compression: Techniques like "gisting" or "Light Thinker" compress prompts into learned embeddings to save context. While efficient, these methods focus on summarization rather than the strategic removal of erroneous reasoning steps.

RL for Reasoning: Recent studies indicate that cognitive behaviors such as verification and backtracking significantly enhance RL training for reasoning. However, current literature overlooks the accumulation of redundant information. Our work extends this by facilitating dynamic context management directly through the RL policy.

2 Problem Formulation and Proposed Work

2.1 Problem Formulation (Finite MDP)

We formalize the reasoning process as a **Finite Markov Decision Process (MDP)**, denoted by the tuple $\mathcal{M} = \langle S, A, P, R \rangle$:

- S : The set of states, where a state s_t represents the prompt and generated context history.

- A : The set of actions. This includes the standard vocabulary $\mathcal{V}$ and special control tokens: $A = \mathcal{V} \cup \{\langle \text{think} \rangle, \langle \text{answer} \rangle, \langle \text{clean} \rangle\}$.
- $P : S \times A \times S \rightarrow [0, 1]$: The probability of transition between states.
- $R : S \times A \times S \rightarrow \mathbb{R}$: The expected reward function.
- $\pi : S \times A \rightarrow [0, 1]$: The policy (the language model) we aim to optimize.

The objective is to learn an *optimal policy* π^* that maximizes the total expected reward:

$$\max_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)] \quad (1)$$

where the trajectory τ may include actions that reset the state history via the $\langle \text{clean} \rangle$ token.

2.2 Proposed Methodology

We propose replicating and extending the "Context Cleaning" framework using the Qwen-2.5 0.5B model.

1. Data Curation (SFT): We will curate an SFT dataset based on expert traces. For correct traces, we retain the standard format. For incorrect/unproductive traces, we augment the data to include a realization of error followed by the cleaning token:

`<think> ... [unproductive search] ... Let me reset. </think> <clean>`

This explicitly teaches the model to recognize flawed reasoning and recover.

2. Context Manager Implementation: We will implement a middleware "Context Manager" (Figure 1 in architecture). During inference, if the model generates $\langle \text{clean} \rangle$, the manager halts generation, clears the context history y^0 , and re-prompts the model for a fresh attempt y^1 .

3. Modified RLOO Optimization: We will implement the RLOO algorithm adapted for context management. The advantage for a trajectory involving a clean is computed by considering the final reward of the *entire* interaction. The gradient update is modified to:

$$\nabla J(\theta) \approx \frac{1}{k + k_c} \sum_{i=1}^k A(x, y_i) (\nabla \log \pi(y_i^0 | x) + \mathbb{I}[y_i^1 \neq \emptyset] \cdot \nabla \log \pi(y_i^1 | x)) \quad (2)$$

This ensures the cleaning signal backpropagates through both the decision to clean (y^0) and the subsequent answer (y^1).

4. Future Extensions: While the baseline allows single-use cleaning, we propose extending the MDP to support **Multi-Step Cleaning** and **Selective Memory Retention**. This involves modifying the Context Manager to allow $k > 1$ resets, penalized by a step-cost in the reward function to prevent infinite loops, enabling recovery from deeper reasoning errors.

3 Proposed Evaluation

Dataset: We will use the **Countdown dataset**, specifically focusing on "Hard" problems (involving multiplication/division) where context clutter is most detrimental.

Baselines:

1. **Default SFT/RLOO:** Standard training without the $\langle \text{clean} \rangle$ token mechanism.
2. **Extension SFT:** The model fine-tuned to clean but without RL optimization.

Metrics:

- **Accuracy:** We aim to replicate the improvement on hard problems (targeting an increase from $\approx 13\%$ to $\approx 36\%$).

- **Clean Efficiency:** We will measure the usage rate of the <clean> token. We hypothesize that RL training will reduce the frequency of cleaning compared to SFT, while increasing the average reward of cleaned trajectories (indicating "judicious" usage).

References

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